

```
import cv2
```

Reading images using imread() function

```
img = cv2.imread(r"test.jpg")  
print(img)
```

Output

```
[[[ 76  44   3]  
   [ 68  37   0]  
   [ 66  34   0]  
   ...  
   [  6  85  66]  
   [  0  46  29]  
   [ 17  60  45]]  
  
[[ 79  46   1]  
 [ 73  41   0]  
 [ 70  39   0]  
 ...  
 [ 60 141 122]  
 [ 46 106  88]  
 [ 60 109  93]]  
  
[[ 89  56   6]  
 [ 87  54   5]  
 [ 83  53   6]  
 ...  
 [ 84 173 153]  
 [ 56 127 110]  
 [ 64 127 111]]  
  
...  
  
[[  3  46   3]  
 [  0  34   0]  
 [  0  37   0]  
 ...  
 [ 14  18  12]  
 [  0   5   0]  
 [ 19  27  17]]  
  
[[ 11  47  11]  
 [  0  33   0]  
 [  0  34   0]  
 ...  
 [ 15  19  13]  
 [ 17  25  15]  
 [ 17  25  15]]  
  
[[ 21  54  20]  
 [  4  37   3]  
 [  5  37   6]  
 ...  
 [ 78  82  76]  
 [ 89  97  87]  
 [ 44  52  42]]]
```

= Getting image height and width

```
img = cv2.imread(r"test.jpg")
# print(img.shape[:2])
print(f"Height: {img.shape[0]} Width: {img.shape[1]}")
```

Output

Height: 1000 Width: 1600

= Displaying Image

```
cv2.imshow("image", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Extracting RGB Values

```
B, G, R = img[100][100]
print(f"Red: {R}\nGreen: {G}\nBlue: {B}")
```

Output

Red: 4

Green: 38

Blue: 75

```
cv2.imshow("image", img[100][100])
cv2.waitKey(0)
cv2.destroyAllWindows()
```

```
B = img[100][100][0]
G = img[100][100][1]
R = img[100][100][2]
print(f"Red: {R}\nGreen: {G}\nBlue: {B}")
```

Output

Red: 4

Green: 38

Blue: 75

= Grayscale images

```
img = cv2.imread("gray_original.png")
cv2.imshow("Original", img)
cv2.waitKey(0)
cv2.destroyAllWindows()

gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2.imshow("Grayscale", img)
cv2.waitKey(0)
cv2.destroyAllWindows()

cv2.imwrite("gray_greyscale.png", gray_img)
```

Output

True

= BGR2RGB

```
img = cv2.imread("gray_original.png", 1)
```

```
cv2.imshow("Original", img)
cv2.waitKey(0)
cv2.destroyAllWindows()

gray_img = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
cv2.imshow("Grayscale", gray_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

BGR2HSV

```
img = cv2.imread("test.jpg")
cv2.imshow("Original", img)
cv2.waitKey(0)
cv2.destroyAllWindows()

hsv_img = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
cv2.imshow("HSV", hsv_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Splitting R, G, B

```
img = cv2.imread("test.jpg")
B, G, R = cv2.split(img)
cv2.imshow("Red", R)
cv2.waitKey(0)
cv2.destroyAllWindows()

cv2.imshow("Blue", B)
cv2.waitKey(0)
cv2.destroyAllWindows()

cv2.imshow("Green", G)
cv2.waitKey(0)
cv2.destroyAllWindows()
```